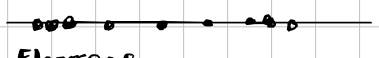
Plasmons

A small disturbance in electron density produces an electric field, which drives the electrons back toward equilibrium
Because electrons have inertia, they overshoot equilibrium and continue oscillating
This oscillation is very different from ordinary electromagnetic wave.
There are two types of waves

- Transverse EM wave



$\vec{E} \perp \vec{K}$



$\vec{E} \parallel \vec{K}$

For a transverse electromagnetic wave in a homogeneous medium

$$\vec{E} \perp \vec{K}$$

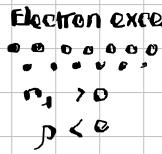
for a longitudinal plasma wave

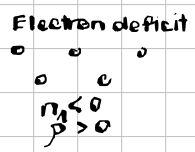
$$\vec{E} \parallel \vec{K}$$

Where does the density wave come from? Imagine a metal containing many mobile electrons and a fixed positive ionic background
At equilibrium, the positive charge density is en_0 and the negative charge density is $-en_0$
Suppose the electron density is disturbed

$$n(x,t) = n_0 + n_1(x,t) \qquad n_1(x) \propto \sin(qx)$$

Some regions contain more electrons and some contain less electrons than average





Electron density and charge density have opposite signs.

Therefore

$$\rho(x) = en_0 - e(n_0 + n_1(x)) = -en_1(x)$$

Density creates a restoring force. From Gauss's law

$$\nabla \cdot \vec{E} = 4\pi\rho$$

Because oscillation is along x

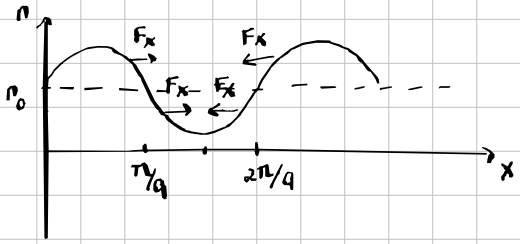
$$\frac{\partial E_x}{\partial x} = 4\pi\rho(x)$$

We know $\rho(x) = -eA\sin(qx)$ where A is the amplitude of the electron-density fluctuation

$$\frac{\partial E_x}{\partial x} = -4\pi e A \sin(qx) \Rightarrow E_x(x) = \frac{4\pi eA}{q}\cos(qx)$$

The force on the electron is

$$F_x = -eE_x \Rightarrow F_x(x) = -\frac{4\pi e^2 A}{q}\cos qx$$





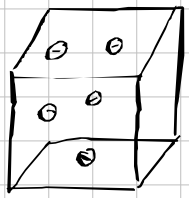


The force is zero exactly at the center of a density peak.

Consider Newton's equation for electron i

$$m\frac{d\vec{v}_i}{dt} = -e\vec{E}(\vec{r}_i)$$ m: electron's mass $\vec{v}_i$: velocity of electron E_i: Electric field

Areal plasma contains an enormous number of electrons in a volume



Define local electron number density

$$n(r) = \frac{\Delta N}{\Delta V}$$

Define average electron velocity:

$$\vec{V}(\vec{r}) = \frac{1}{\Delta N}\sum_i \vec{v}_i$$

The electric current density is $\vec{J}_m = -\frac{e}{\Delta V}\sum_i \vec{v}_i$

Differentiate

$$\frac{d\vec{J}_m}{dt} = -\frac{e}{\Delta V}\sum_i \frac{d\vec{v}_i}{dt} = \frac{e^2}{m\Delta V}\sum_i \vec{E}(\vec{r}_i)$$

Assume that $\vec{E}(\vec{r}_i) \approx \vec{E}(\vec{r})$

$$\frac{d\vec{J}_m}{dt} = \frac{e^2}{m}\frac{\Delta N}{\Delta V}\vec{E} \qquad \vec{E} = \frac{1}{\Delta N}\sum_i \vec{E}(\vec{r}_i)$$

For small fluctuations around the equilibrium density $n(\vec{r},t) \approx n_0$

Then

$$\frac{\partial \vec{J}_m}{dt} = \frac{e^2}{m} n_0 \vec{E}$$

Using the continuity equation $\nabla\cdot\vec{J}_m = -\frac{\partial \rho_m}{\partial t}$

That is

$$\frac{\partial \rho_m}{\partial t} + \nabla\cdot\vec{J}_m = 0 \Rightarrow \frac{\partial^2}{\partial t^2}\rho_m = -\frac{\partial}{\partial t}(\nabla\cdot\vec{J}_m) \Rightarrow \frac{\partial^2}{\partial t^2}\rho_m = -\frac{e^2 n_0}{m}\nabla\cdot\vec{E}$$

Using $\nabla\cdot\vec{E} = 4\pi\rho_m$, we obtain

$$\frac{\partial^2 \rho_m}{\partial t^2} + \frac{4\pi e^2 n_0}{m}\rho_m = 0$$

By defining $\omega_{pe}^2 = \frac{4\pi e^2 n_0}{m}$

$$\frac{\partial^2}{\partial t^2}\rho_m + \omega_{pe}^2\rho_m = 0 \qquad \rho_m(\vec{r},t) = \rho_m(\vec{r},0)\cos(\omega_{pe}t)$$